

FARM MANAGEMENT SYSTEM

PROJECT DOCUMENT

PROJECT TEAM MEMBERS

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FARM MANAGEMENT SYSTEM

Project Documentation

1. Abstract

The Farm Management System is a web-based application designed to help farmers manage and monitor their farming activities efficiently. Traditional farming management often involves maintaining records manually, which can be time-consuming and may lead to errors or loss of information. The proposed system provides a centralized platform for managing farm-related information such as crops, fields, farming activities, expenses, workers, and harvest details. It helps farmers maintain accurate records and access information easily.

2. Problem Statement

Farmers often maintain farm records manually using notebooks or separate files. This makes it difficult to track crop information, field details, farming activities, expenses, workers, and harvest quantities. Manual record keeping can result in data loss, calculation errors, and difficulty in accessing previous records. Therefore, a computerized Farm Management System is proposed to manage farm activities in an organized and efficient manner.

3. Objectives

- Maintain farm and field information digitally.
- Manage crop details efficiently.
- Record daily farming activities.
- Maintain records of farm expenses.
- Manage worker information.
- Record harvest and production details.
- Reduce manual paperwork.
- Provide quick access to farm records.
- Improve farm-management efficiency.
- Generate useful reports for decision-making.

4. Project Scope

The Farm Management System can be used to manage different aspects of agricultural operations. It includes farmer/admin login, farm management, field management, crop management, activity management, expense management, worker management, harvest management, dashboard, and reports. Future versions can include weather forecasting, IoT sensors, AI-based crop recommendations, and market-price information.

5. Existing System

In the existing system, many farmers manage farm information manually. This is time-consuming, paper-based, difficult to search, vulnerable to data loss and calculation errors, and makes report generation difficult. There is also no centralized database for monitoring overall farm performance.

6. Proposed System

The proposed Farm Management System provides a digital platform to manage farm-related activities. The administrator or farmer can add, update, delete, and view information related to crops, fields, workers, expenses, and harvests. A dashboard provides an overview of important farm information, making it easier to monitor farming operations.

7. Modules

7.1 User Login Module: Provides secure access through login, logout, authentication, and password management.

7.2 Farm Management: Stores farm name, location, area, soil type, and owner details.

7.3 Field Management: Maintains field name/number, area, soil type, irrigation method, and current crop.

7.4 Crop Management: Manages crop name, type, planting date, expected harvest date, field, quantity planted, and status.

7.5 Farming Activity: Records ploughing, sowing, irrigation, fertilizing, spraying, weeding, and harvesting activities.

7.6 Expense Management: Records seeds, fertilizers, pesticides, labour, equipment, irrigation, and transportation expenses.

7.7 Worker Management: Maintains worker name, contact, work type, joining date, salary/wage, and status.

7.8 Harvest Management: Records crop, harvest date, quantity, quality, selling price, and revenue.

7.9 Dashboard: Displays total fields, active crops, expenses, workers, harvest quantity, and revenue.

7.10 Reports: Generates crop, expense, harvest, worker, and activity reports.

8. System Requirements

Hardware: Computer/Laptop, minimum 4 GB RAM, minimum 20 GB available storage, keyboard and mouse, and internet connection.

Software: Windows/Linux, HTML, CSS, JavaScript/React, Node.js/PHP, MySQL, Chrome/Edge, and Visual Studio Code. Technologies can be adjusted according to the actual implementation.

9. Technologies Used

HTML – Web page structure

CSS – Styling and layout

JavaScript – Interactivity

React – Frontend development

Node.js – Backend development

MySQL – Database management

Visual Studio Code – Development environment

10. Application Flow

START → User Login → Dashboard → Farm Management / Crop Management / Field Management → Farming Activities → Expense Management → Worker Management → Harvest Management → Reports → Logout → END

11. Database Design

The database may contain User, Farm, Crop, Expense, Worker, and Harvest tables.

User: user_id, name, email, password.

Farm: farm_id, farm_name, location, area, soil_type.

Crop: crop_id, crop_name, field_id, planting_date, harvest_date, status.

Expense: expense_id, category, amount, date, description.

Worker: worker_id, name, contact, work_type, wage.

Harvest: harvest_id, crop_id, harvest_date, quantity, price, revenue.

12. Functional Requirements

The system should allow users to register/login, add farm details, manage fields, add crop information, update crop status, record farming activities, manage expenses and workers, record harvest details, view farm statistics, generate reports, and update/delete records.

13. Non-Functional Requirements

Performance – quick response to user requests.

Security – protect user information and credentials.

Usability – simple and easy-to-understand interface.

Reliability – accurate storage of records.

Scalability – support additional farms, crops, workers, and records.

14. Advantages

Reduces paperwork; saves time; provides centralized data management; makes records easy to access; reduces calculation errors; helps monitor crop activities; simplifies expense tracking; maintains harvest records; provides useful reports; and improves farm management.

15. Future Enhancements

Weather forecasting, IoT integration for soil moisture/temperature/humidity monitoring, AI crop recommendations, AI-based disease detection, market-price information, mobile application support, online notifications, and GPS-based farm/field tracking.

16. Expected Output

The final system should provide a user-friendly dashboard where farmers can manage farm information, fields, crops, farming activities, expenses, workers, and harvests, while viewing statistics and generating reports.

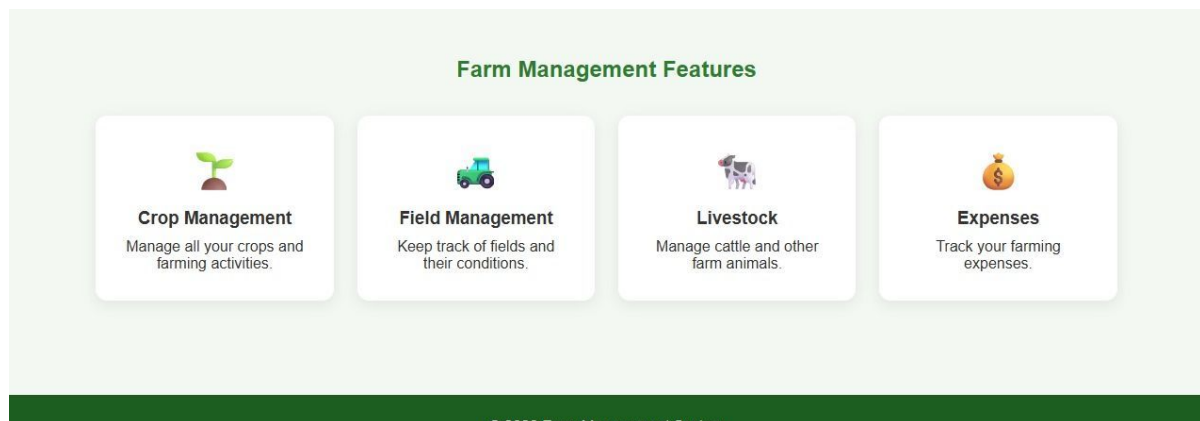
17. Challenges Faced

Possible challenges include designing the database structure, connecting frontend and backend, implementing authentication, managing multiple farm records, calculating expenses and revenue, designing a simple user interface, and maintaining data consistency.

18. Learning Outcomes

Students gain knowledge in web application development, database management, frontend and backend development, CRUD operations, authentication, data visualization, project planning, software testing, and project documentation.

PROJECT SCREENSHOTS





Welcome to My Farm

Manage your complete farm from this dashboard.

 Crops**0**

Total Crops

 Fields**0**

Total Fields

 Livestock**0**

Total Animals

 Expenses**₹0**

Total Expenses

Farm Management

 Crop Management[← Dashboard](#)

Crop Management

Add and manage your farm crops.

[Add Crop](#)

Crop

Field

Sowing Date

Action

 Livestock Management

Manage your farm animals.

[Add Livestock](#)**Animal****Quantity****Action** **Expense Management**

Track all your farming expenses.

[Add Expense](#)**Expense****Amount****Action**



Harvest Management

Record your harvested crops.

Add Harvest

Crop	Quantity	Action
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Sales Management

Record your farm sales.

Add Sale

Crop	Quantity	Amount	Action
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Farm Reports

Overview of your farm.

 Crops

0

 Fields

0

 Animals

0

 Expenses

₹0

 Sales

₹0

20. Conclusion

The Farm Management System provides a digital solution for managing various farming activities. It helps farmers maintain organized records of farms, fields, crops, workers, expenses, activities, and harvests.

By reducing manual paperwork and providing centralized access to information, the system can make farm management more organized and efficient. The project can also be extended with modern technologies such as AI, IoT, weather forecasting, and mobile applications to provide more advanced agricultural support.

